

Electric Field of Cylinder by Elementary Integration

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§1 Introduction and Problem Set-Up

Calculating the electric field produced by various distributions of charge is a common problem in introductory physics. Students begin by confronting the electric field produced by point charges and then build up to more complicated distributions like a uniformly charged disk or line segment. To find the electric fields of these shapes, the method of superposition of component electric fields is introduced, where the function for the electric field produced by an infinitesimal piece of charge is integrated over the entire shape.

For more complicated shapes, Gauss's Law is used to simplify the calculation of the electric field. In the present paper, we describe how to calculate the electric field on the surface of a uniformly charged, infinitely long cylinder, which is approximated by cylinders where the length is far larger than the radius. We show the common way of finding the field by Gauss's Law, but then arrive at the same answer by the less common method of superposition of component fields. Although this method is more complicated, it is instructive to be able to solve a problem with independent strategies. We also appreciate how elementary tools that most people who have taken a calculus class will understand can approach complex problems and beautifully arrive at elegant answers.

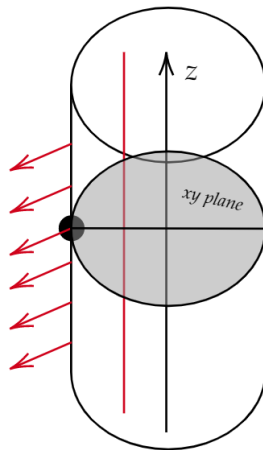


Figure 1: 3D view of cylindrical charge distribution

In Figure 1 above, we orient our cylinder with respect to the z axis. Note that due to the symmetry of the cylinder, all points on the surface are equivalent and the electric field will be uniform. Each point “sees” the same types of charges; the infinite length of the cylinder makes the z coordinate of the point irrelevant and the symmetry of a circle makes the position on the circle independent of the field. Thus, we can set the point of interest to the origin. We assume the cylinder is positively charged. Also due to symmetry, the direction of the field must be radially outward. For every point in the cylinder, there is a corresponding point that has an

opposite y and z coordinate, which has an equal contribution to the electric field as the first point, except with opposite y and z components. Therefore, the y and z components cancel and we are left with x components to add up, leading to a total electric field that is radially outward since electric fields point away from positive charges.

Instead of integrating the contributions of each point charge in the cylinder, it was more simple to envision the cylinder as made up of infinitely long wires. We then integrate the contributions of each of these very thin and long wires, reducing a triple integral to a double integral. One of these contributions is shown in Figure 1.

§2 The Electric Field by Gauss's Law

The easiest method to solve the problem is to invoke Gauss's Law, which states that the surface integral of the electric field (i.e. the flux, Φ) on a closed surface is equal to the total charge, Q enclosed by the surface divided by the vacuum permittivity, ϵ_0 . To apply Gauss's Law to the cylinder, we need a closed surface. The infinite cylinder has radius R . Envision a finite cylinder of length L and radius R . The constant volume charge density is ρ and the surface area of the sides of the cylinder is A . Since the electric field is uniform, we can express the flux as the product of the surface area and the electric field. This leads to:

$$\begin{aligned}\Phi = Q/\epsilon_0 = \rho V/\epsilon_0 &= \frac{\rho(\pi R^2 L)}{\epsilon_0} \implies \Phi = EA = E(2\pi RL) = \frac{\rho(\pi R^2 L)}{\epsilon_0} \\ \implies E &= \boxed{\frac{\rho R}{2\epsilon_0}}.\end{aligned}$$

§3 The Electric Field by Superposition of Component Electric Fields

§3.1 Field Produced by an Uniformly Charged Infinite Wire

Set the point of interest and the infinite wire in the ab plane with the wire on the b axis. For every point on an infinite wire, there is a corresponding point whose b coordinate is opposite. The magnitude of the contributions of each of these points are equal, but the b components cancel due to their opposite direction. Thus, we only care about the a component. Let α be the angle that the electric field produced by a infinitesimal piece of charge, dQ , makes with the a axis. For future reference, a will act as the distance of the point of interest to the wire.

Applying Coulomb's Law, expressing the charge in terms of the linear charge density, λ , and expressing the cosine in terms of a and b using geometry, we get:

$$\begin{aligned}E_{wire} &= \int dE_a = \int k \frac{dQ}{d^2} \cos \alpha = \int \frac{k\lambda db}{a^2 + b^2} \frac{a}{\sqrt{a^2 + b^2}} \\ &= k\lambda a \int_{-\infty}^{\infty} \frac{1}{(a^2 + b^2)^{3/2}} db.\end{aligned}$$

The sum of squares, $a^2 + b^2$, in the denominator suggests a tangent substitution. Let $b = a \tan u$. After adjusting the bounds and applying $db = a \sec^2 u du$,

$$E_{wire} = \int dE_a = k\lambda a \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{a \sec^2 u}{(a^2(1 + \tan^2 u))^{3/2}} du.$$

Using the trigonometric identity $1 + \tan^2 u = \sec^2 u$, the expression simplifies nicely to:

$$E_{wire} = \int dE_a = k\lambda a \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{a^2 \sec u} du = k\lambda a \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos u}{a^2} du = \frac{2k\lambda}{a} = \boxed{\frac{\lambda}{2\pi\epsilon_0 a}}.$$

§3.2 Superposition of the Electric Fields Produced by Infinite Wires

Now that we have the electric field produced by each component wire, we must integrate over the entire cylinder, i.e. the disk shown below in Figure 2.

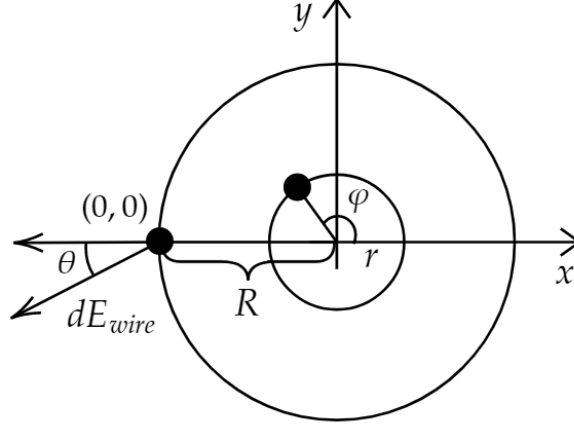


Figure 2: 2D view of electric field contribution by point in disk

Again, due to symmetry, the y contributions of each wire are cancelled by a corresponding wire that has an opposite y coordinate. So we integrate the x components of the electric field over the entire disk.

$$E_{cylinder} = \iint dE_{wire,x} = \iint \frac{\lambda}{2\pi\epsilon_0 a} \cos \theta dA.$$

By the geometry of the setup, $a = \sqrt{x^2 + y^2}$ and $\cos \theta = \frac{x}{\sqrt{x^2 + y^2}}$, which gives:

$$E_{cylinder} = \iint dE_{wire,x} = \iint \frac{\lambda}{2\pi\epsilon_0} \frac{x}{x^2 + y^2} dy dx.$$

We will name the constant on the left A , so that

$$A = \frac{\lambda}{2\pi\epsilon_0}.$$

By converting to polar coordinates using the parametrization

$$x = R + r \cos(\varphi)$$

$$y = r \sin(\varphi),$$

we obtain

$$E_{cylinder} = A \iint \frac{(R + r \cos \varphi)r}{r^2 + 2R \cos(\varphi)r + R^2} dr d\varphi,$$

where we substituted the Jacobian determinant associated with the change in variables to polar coordinates, i.e. $dy dx = r dr d\varphi$.

We observe that there is an irreducible quadratic in the denominator, suggesting the method of completing the square and eventually doing a tangent substitution. When we complete the square and include the bounds for r , we obtain:

$$E_{cylinder} = A \int_0^R \int_0^{2\pi} \frac{(R + r \cos(\varphi))r}{(r + R \cos \varphi)^2 - R^2 \cos^2 \varphi + R^2} dr d\varphi = A \int_0^R \int_0^{2\pi} \frac{(R + r \cos \varphi)r}{(r + R \cos \varphi)^2 + R^2(1 - \cos^2 \varphi)} dr d\varphi$$

$$= A \iint_0^R \frac{r^2 \cos \varphi + Rr}{(r + R \cos \varphi)^2 + R^2 \sin^2 \varphi} dr d\varphi.$$

Before we perform the tangent substitution, we must first simplify the expression so that the bottom is a sum of squares of single variables. Since we are integrating with respect to r , we can first simplify our expression by defining $T = R \sin \varphi$. For the substitution, let $u = r + R \cos \varphi$. Since $du = dr$, we get:

$$\begin{aligned} E_{cylinder} &= A \iint_{R \cos \varphi}^{R(1+\cos \varphi)} \frac{(u - R \cos \varphi)^2 \cos \varphi + R(u - R \cos \varphi)}{u^2 + T^2} du d\varphi \\ &= A \iint_{R \cos \varphi}^{R(1+\cos \varphi)} \frac{\cos \varphi u^2 - 2R^2 \cos^2 \varphi u + (R^2 \cos^3 \varphi + Ru - R^2 \cos \varphi)}{u^2 + T^2} du d\varphi \\ &= A \iint_{R \cos \varphi}^{R(1+\cos \varphi)} \frac{\cos \varphi u^2 - (2R^2 \cos^2 \varphi - R)u + R^2(\cos^3 \varphi - \cos \varphi)}{u^2 + T^2} du d\varphi. \end{aligned}$$

We now have two polynomials in u of degree 2, meaning we can do long division and obtain a constant plus a fraction with the numerator as a polynomial in u of degree strictly less than 2.

$$\begin{aligned} E_{cylinder} &= A \iint_{R \cos \varphi}^{R(1+\cos \varphi)} \cos \varphi + \frac{(R - 2R \cos^2 \varphi)u + (R^2(\cos^3 \varphi - \cos \varphi) - T^2 \cos \varphi)}{u^2 + T^2} du d\varphi \\ &= A \iint_{R \cos \varphi}^{R(1+\cos \varphi)} \cos \varphi + \frac{(R - 2R \cos^2 \varphi)u + R^2(\cos^3 \varphi - \cos \varphi - \sin^2 \varphi \cos \varphi)}{u^2 + T^2} du d\varphi. \end{aligned}$$

Breaking down the integral of the sum of the integrals, $E_{cylinder} = A(I_1 + I_2)$, where

$$\begin{aligned} I_1 &= \iint_{R \cos \varphi}^{R(1+\cos \varphi)} \cos \varphi du d\varphi, \\ I_2 &= \iint_{R \cos \varphi}^{R(1+\cos \varphi)} \frac{(R - 2R \cos^2 \varphi)u + R^2(\cos^3 \varphi - \cos \varphi - \sin^2 \varphi \cos \varphi)}{u^2 + T^2} du d\varphi. \end{aligned}$$

Let us evaluate I_1 first. This is easy as $\cos \varphi$ is a constant with respect to u .

$$I_1 = \iint_{R \cos \varphi}^{R(1+\cos \varphi)} \cos \varphi du d\varphi = \int \cos \varphi \cdot (R(1 + \cos \varphi) - R \cos \varphi) d\varphi = \int R \cos \varphi d\varphi.$$

To evaluate I_2 , we should split it up into two more integrals $I_2 = I_3 + I_4$, with

$$\begin{aligned} I_3 &= \iint_{R \cos \varphi}^{R(1+\cos \varphi)} \frac{(R - 2R \cos^2 \varphi)u}{u^2 + T^2} du d\varphi, \\ I_4 &= \iint_{R \cos \varphi}^{R(1+\cos \varphi)} \frac{R^2(\cos^3 \varphi - \cos \varphi - \sin^2 \varphi \cos \varphi)}{u^2 + T^2} du d\varphi. \end{aligned}$$

For I_3 ,

$$I_3 = \iint_{R \cos \varphi}^{R(1+\cos \varphi)} \frac{(R - 2R \cos^2 \varphi)u}{u^2 + T^2} du d\varphi = \frac{R - 2R \cos^2 \varphi}{2} \iint_{R \cos \varphi}^{R(1+\cos \varphi)} \frac{2u}{u^2 + T^2} du d\varphi.$$

The multiplication by 2 in numerator and denominator sets up a nice substitution of $v = u^2 + T^2$, with $dv = 2u du$. This substitution gives

$$I_3 = \frac{R - 2R \cos^2 \varphi}{2} \iint_{R^2}^{2R^2(1+\cos \varphi)} \frac{1}{v} dv d\varphi.$$

We finally arrive at the point where we can take an elementary integral with ease, giving us:

$$\begin{aligned} I_3 &= \int \frac{R - 2R \cos^2 \varphi}{2} \ln \left(\frac{2R^2(1 + \cos \varphi)}{R^2} \right) d\varphi = \int \frac{R - 2R \cos^2 \varphi}{2} \ln(2(1 + \cos \varphi)) d\varphi \\ &= \int -\frac{R \cos(2\varphi)}{2} \ln(2(1 + \cos \varphi)) d\varphi. \end{aligned}$$

Approaching I_4 , we will simplify the constant in the numerator and then finally perform the long awaited tangent substitution.

$$\begin{aligned} I_4 &= \iint_{R \cos \varphi}^{R(1+\cos \varphi)} \frac{R^2(\cos^3 \varphi - \cos \varphi - \sin^2 \varphi \cos \varphi)}{u^2 + T^2} du d\varphi \\ &= R^2 \cos \varphi (\cos^2 \varphi - \sin^2 \varphi - 1) \iint_{R \cos \varphi}^{R(1+\cos \varphi)} \frac{1}{u^2 + T^2} du d\varphi \\ &= R^2 \cos \varphi (\cos(2\varphi) - 1) \iint_{R \cos \varphi}^{R(1+\cos \varphi)} \frac{1}{u^2 + T^2} du d\varphi. \end{aligned}$$

To simplify the expressions for now, let $B = R^2 \cos \varphi (\cos(2\varphi) - 1)$. To perform the tangent substitution, let $u = T \tan \beta$, so that $du = T \sec^2 \beta d\beta$. This gives us:

$$\begin{aligned} I_4 &= B \iint_{R \cos \varphi}^{R(1+\cos \varphi)} \frac{1}{u^2 + T^2} du d\varphi = B \int \int_{\tan^{-1} \frac{R \cos \varphi}{T}}^{\tan^{-1} \frac{R(1+\cos \varphi)}{T}} \frac{T \sec^2 \beta}{T^2(\tan^2 \beta + 1)} d\beta d\varphi \\ &= B \int \int_{\tan^{-1} \frac{R \cos \varphi}{T}}^{\tan^{-1} \frac{R(1+\cos \varphi)}{T}} \frac{1}{T} d\beta d\varphi \\ &= \int \frac{B}{T} \left(\tan^{-1} \frac{R(1+\cos \varphi)}{T} - \tan^{-1} \frac{R \cos \varphi}{T} \right) d\varphi. \end{aligned}$$

Note that the B goes back in the integral because integration with respect r and its derived variables is finished so B is no longer a constant. This expression gives us the wonderful opportunity of applying the tangent subtraction formula. If we let

$$\begin{aligned} c &= \tan^{-1} \frac{R(1+\cos \varphi)}{T}, \\ d &= \tan^{-1} \frac{R \cos \varphi}{T}, \end{aligned}$$

then by the tangent subtraction formula,

$$\tan(c - d) = \frac{\tan c - \tan d}{1 + \tan c \tan d} = \frac{\frac{R}{T}}{1 + \frac{R^2 \cos \varphi (1 + \cos \varphi)}{T^2}} = \frac{RT}{T^2 + R^2 \cos \varphi (1 + \cos \varphi)}.$$

The inverse tangent terms in I_4 are equal to $(c - d)$, so by taking the inverse tangent of the equation found by the tangent subtraction formula,

$$c - d = \tan^{-1} \frac{RT}{T^2 + R^2 \cos \varphi (1 + \cos \varphi)}.$$

Therefore,

$$I_4 = \int \frac{B}{T} \tan^{-1} \frac{RT}{T^2 + R^2 \cos \varphi (1 + \cos \varphi)} d\varphi.$$

Now we can plug back in the expressions for the constants that we previously defined and solve our four integrals with respect to φ . The bounds for φ are from 0 to 2π given that we are integrating over the entire cylinder.

$$I_1 = \int_0^{2\pi} R \cos \varphi d\varphi = R(\sin 2\pi - \sin 0) = 0;$$

We see the first integral quickly disappears.

$$\begin{aligned} I_3 &= \int_0^{2\pi} -\frac{R \cos(2\varphi)}{2} \ln 2(1 + \cos \varphi) d\varphi = \int_0^{2\pi} -\frac{R \cos(2\varphi)}{2} (\ln(1 + \cos \varphi) + \ln 2) d\varphi \\ &= -\frac{R \ln 2}{2} \int_0^{2\pi} \cos(2\varphi) d\varphi - \int_0^{2\pi} \frac{R \cos(2\varphi)}{2} \ln(1 + \cos \varphi) d\varphi. \end{aligned}$$

The first integral of the previous line evaluates to 0 conveniently, using the elementary integral of $\cos(2\varphi)$. Now evaluating the second integral in the expression by parts, let $u = \ln(1 + \cos \varphi)$, $dv = \frac{R \cos(2\varphi)}{2} d\varphi$, so that $du = -\frac{\sin \varphi}{1 + \cos \varphi} d\varphi$ and $v = \frac{R \sin(2\varphi)}{4}$. Applying integration by parts,

$$\begin{aligned} I_3 &= - \int_0^{2\pi} \frac{R \cos(2\varphi)}{2} \ln(1 + \cos \varphi) d\varphi \\ &= \left(-\frac{\ln(1 + \cos(2\pi)) R \sin(4\pi)}{4} + \frac{\ln(1 + \cos 0) R \sin 0}{4} \right) + \int_0^{2\pi} -\frac{R \sin(2\varphi)}{4} \frac{\sin \varphi}{1 + \cos \varphi} d\varphi \\ &= - \int_0^{2\pi} \frac{R \sin(2\varphi)}{4} \frac{\sin \varphi}{1 + \cos \varphi} d\varphi = - \int_0^{2\pi} \frac{R \sin^2 \varphi \cos \varphi}{2(1 + \cos \varphi)} d\varphi \\ &= -\frac{R}{2} \int_0^{2\pi} \frac{\sin^2 \varphi \cos \varphi (1 - \cos \varphi)}{(1 + \cos \varphi)(1 - \cos \varphi)} d\varphi = -\frac{R}{2} \int_0^{2\pi} \cos \varphi (1 - \cos \varphi) d\varphi \\ &= -\frac{R}{2} \left(\int_0^{2\pi} \cos \varphi d\varphi - \int_0^{2\pi} \cos^2 \varphi d\varphi \right). \end{aligned}$$

The integral of cosine from 0 to 2π is 0 and the other integral can be easily shown (say, via symmetry with $\sin^2 \varphi$) to be equal to π , giving a final result for I_3 of

$$I_3 = \frac{\pi R}{2}.$$

To finish the integration of I_4 , we must first plug in the values we defined for B and T :

$$\begin{aligned} I_4 &= \int \frac{B}{T} \tan^{-1} \frac{RT}{T^2 + R^2 \cos \varphi (1 + \cos \varphi)} d\varphi \\ &= \int \frac{R^2 \cos \varphi (\cos(2\varphi) - 1)}{R \sin \varphi} \tan^{-1} \frac{R^2 \sin \varphi}{R^2 \sin^2 \varphi + R^2 \cos \varphi (1 + \cos \varphi)} d\varphi \\ &= \int \frac{R \cos \varphi (-2 \sin^2 \varphi)}{\sin \varphi} \tan^{-1} \frac{\sin \varphi}{\sin^2 \varphi + \cos \varphi (1 + \cos \varphi)} d\varphi \\ &= \int -2R \cos \varphi \sin \varphi \tan^{-1} \frac{\sin \varphi}{1 + \cos \varphi} d\varphi = \int -2R \cos \varphi \sin \varphi \tan^{-1} \tan \frac{\varphi}{2} d\varphi = \int -\frac{R}{2} \sin(2\varphi) \varphi d\varphi. \end{aligned}$$

The key step is invoking the tangent half angle formula in the last step. Our result is an elementary integral that can be found by integrating by parts.

$$I_4 = -\frac{R}{2} \int_0^{2\pi} \sin(2\varphi) \varphi d\varphi = -\frac{R}{2} \left(\frac{\sin(4\pi) - 4\pi \cos(4\pi)}{4} - \frac{\sin(0) - 0 \cos(0)}{4} \right) = \frac{\pi R}{2}.$$

Putting everything together, we arrive at the grand finale of

$$E_{cylinder} = A(I_1 + I_2) = A(I_1 + I_3 + I_4) = A \left(0 + \frac{\pi R}{2} + \frac{\pi R}{2} \right) = A\pi R = \boxed{\frac{\lambda R}{2\epsilon_0}}.$$

The only difference that remains between this formula and the formula found by Gauss's Law is that this one has linear charge density λ , but the Gauss formula uses volume charge density ρ . To show that these two values are equal is trivial. The charge per unit volume can be expressed as

$$dQ = \rho dV = \lambda dz dA = \lambda dx dy dz = \lambda dV,$$

so we conclude that $\rho = \lambda$ and the two formulas for the electric field on the surface of an infinite cylinder are equivalent.

§4 Takeaways and Conclusions

We have successfully shown two methods for finding the electric field on the surface of a uniformly charged, infinitely long cylinder. Gauss's method was far faster, but it relies on the Divergence Theorem, a higher level concept compared to the elementary integration used in the superposition method. Of course, Gauss's method is the method of choice, but going through the fundamental integration that is wrapped up in the Gauss method evokes a sense of wonder at the seemingly magical cancellations that happen to give such a simple answer that is identical to that found by the Gauss method. Highlights of the method include the beautiful reduction of the difference in tangent inverses, the application of the tangent half formula, and surprising ease of equating the linear charge density to the volume charge density. Trigonometric identities abound, this problem was akin to a full body workout for introductory calculus.

No high level concepts were used, but the organization and methodical thinking necessary to parse through the integrals and their sub integrals are certainly skills that are helpful when approaching higher level math. Feel free to try to interpret the integral method physically to point out why and where the right cancellations occur so that we arrive at a clean fraction in the end. It is not trivial! It is ever important to both train one's qualitative skills along with their quantitative ones. This paper shows that there is a wealth of knowledge to be gained by even the simplest of problem statements if we are willing to dig in.